

RAPHAEL ESQUIVEL

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EDUCATION

Harvey Mudd College, Claremont, CA Aug 2025 - May 2029
B.S., Mathematics & Physics GPA: 3.88/4.0
Relevant coursework: Readings in Algebraic Geometry, Numerical Analysis, Abstract Algebra I, Intermediate Probability, Quantum Field Theory, General Relativity

Arizona State University, Phoenix, AZ Aug 2024 - Apr 2025
Non-Degree Seeking Student GPA: 4.0/4.0
Relevant coursework: Mathematical Methods Physics II, Introductory Modern Physics, Quantum Physics II, Classical Particles, Fields, and Matter II

EXPERIENCE

Repair Steward, HMC Makerspace Aug 2026 - Present

Student Researcher, Logarithmic Gromov-Witten Theory (HMC) Nov 2025 - Present

- Analysis of the tropical and combinatorial structure of relative stable maps to $(\mathbb{P}^n, H)$ and their boundary behavior in moduli spaces.

Student Researcher, Quantum Quasars Lab (HMC) Sep 2025 - Present

- Designing and assembling a free-space quantum communication system for a kilometer-scale demonstration of quantum position verification.
- Developed of an ultra-low-latency RF synchronization link and FPGA-based control for RF communication and photonics.
- Built and aligned optical experimental setup and fabricated custom optomechanical components.

AWARDS

Karl Menger Memorial Award (2nd), American Mathematical Society May 2025
Awarded \$1000 at Regeneron ISEF for project *Lie Groups and Algebroids in Lagrangian Reduction*.

SKILLS

Programming languages	Python, Javascript, C/C++, R, Mathematica, Julia, Verilog
Software	COMSOL, Xilinx Vivado, Git, Jupyter
Lab equipment	Free-space optics, lasers, SPAD, Pockels cells, optical fiber
Manufacturing	FDM, SLS, lathe, mill, 3D printer maintenance
Languages	English (native), French (native), Spanish (proficient)

RESEARCH INTERESTS

Algebraic geometry, Gromov-Witten theory, quantum field theory, mirror symmetry